

Estimation: the parameters, and how each is identified

Parent block

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Purpose

This records the implemented specification, the moments intended to inform the parameters, and the distinction between estimated and calibrated quantities. It also sets out the open question about the technology and welfare-weight age profiles. The proposed extensions and robustness exercise below have not been implemented.

1. The model

Preferences

The parents' period utility, with $l_{p,t} = 1 - h_{p,t} - \tau_{p,t}$:

$$U_{p,t} = \phi_1 \frac{c_{p,t}^{1-\rho}}{1-\rho} + \phi_2 \frac{l_{p,t}^{1-\eta}}{1-\eta} + \phi_3 \ln HC_t$$

Parental leisure nets out **both** work and time with the child: each uses one unit of the parent's time budget, while only work earns a wage. The child's period utility, with $l_{c,t} = 1 - \tau_{p,t} - i_{c,t}$:

$$U_{c,t} = \lambda_1 \ln l_{c,t} + \lambda_2 \ln HC_t$$

The family's joint utility, with the parents' welfare weight $\tilde{\mu}_t$:

$$U_{f,t} = \phi_1 \frac{C_t^{1-\rho}}{1-\rho} + \phi_2 \frac{l_{p,t}^{1-\eta}}{1-\eta} + \tilde{\alpha}_1 \ln l_{c,t} + \tilde{\alpha}_2 \ln HC_t$$

$$\tilde{\alpha}_1 = (1 - \tilde{\mu}_t) \lambda_1, \quad \tilde{\alpha}_2 = \tilde{\mu}_t \phi_3 + (1 - \tilde{\mu}_t) \lambda_2$$

So the weight on the child's leisure is set entirely by the child's welfare share, while the weight on skill mixes the parents' valuation ϕ_3 and the child's λ_2 .

The welfare weight is $\tilde{\mu}_t = 1$ for $t < 6$; the child has no separate study choice in those periods. Under the current calibration it declines thereafter:

$$\tilde{\mu}_t = \mu_0 + \mu_1 (t - 5), \quad t \geq 6$$

Human capital

$$HC_{t+1} = R_t \tau_{p,t}^{\sigma_{1,t}} e_{p,t}^{\sigma_{2,t}} HC_t^{\sigma_{3,t}} i_{c,t}^{\sigma_{4,t}}$$

$$\sigma_{j,t} = \exp(\sigma_{j0} + \sigma_{j1}(t - 1)), \quad j = 1, 2, 3$$

$$\sigma_{4,t} = \begin{cases} 0, & t < 6 \\ \exp(\sigma_{40} + \sigma_{41}(t - 5)), & t \geq 6 \end{cases}$$

τ_p is parental time, e_p monetary investment, and i_c the child's own study time. Before $t = 6$, the implemented production function omits the study-time factor. From $t = 6$, its elasticity uses $(t - 5)$, so the first active value is $\exp(\sigma_{40} + \sigma_{41})$.

The zero restriction applies to the early entries of `sigma_4_vector`, not to σ_{41} . The code treats t as child age, solves ages 1–17, and activates study at age 6: “zero through age 6 inclusive” would describe a different boundary. There is no attribution to Sahber of a restriction $\sigma_{41} = 0$.

Currently $\sigma_{41} = 0.02$, so the elasticity rises after activation. Self-productivity is calibrated at $\sigma_{3,t} = \exp(-0.90) \approx 0.407$.

Budget

$$a_{t+1} = (1 + r)a_t + y + \lambda(w_t h_{p,t})^{1-\tau} - c_{p,t} - e_{p,t}, \quad a_{t+1} \geq 0$$

with the HSV/Benabou progressive tax; τ is progressivity and λ the level.

Terminal value at separation

$$V_p^{\text{term}} = \psi \ln k + \kappa_{\text{term}} \ln a$$

where k is the child's human capital at 18 and a the parents' retained assets. This is the parents' terminal component. The full continuation value also incorporates the child's lifecycle value, the transfer decision, and the college/work comparison; it is not just the expression above.

2. What changed

The preference weights are time-invariant. $\phi_1, \phi_2, \phi_3, \lambda_1, \lambda_2$ were age profiles $\phi_{j0} + \phi_{j1}(t - 1)$; they are now scalars.

$\phi_1 = 1$ and $\lambda_1 = 1$. These are the chosen normalisations in the current specification. Neither is estimated.

$\psi = 0$. The parents no longer place a separate terminal weight on the child's skill; they value it through the flow term $\phi_3 \ln HC_t$ and through altruism.

Initial human capital comes from the data. HC_1 was an arbitrary $U(0, 1)$ draw. The model's human capital is now measured in the units of the Woodcock–Johnson PCA composite, and the initial draw is taken from the data: the composite is not administered before age 3, so the mean and standard deviation of $\ln HC$ are fitted log-linearly on ages 3–17 and extrapolated back, giving $\ln HC_1 \sim N(5.953, 0.067^2)$ — a median of approximately 385 and a 10–90 range of approximately 353–419.

The timing issue is resolved. The fit is now evaluated at **child age 1**, which is the child age of the first simulation column: the family stage runs $t = 1 \dots 17$ over child ages 1–17 and has no age-0 period. The code previously placed the age-0 extrapolation, $N(5.929, 0.070^2)$, in that column, understating initial skill by 0.024 log points, or 2.4% in levels.

3. The parameters and the moments intended to identify them

Parameter	Role	Main informative moments / current status
ϕ_1	weight on consumption	fixed at 1 (normalisation)
ϕ_2	weight on parental leisure	mean hours worked
ϕ_3	parents' weight on the child's skill	parental time and monetary investment
λ_1	child's weight on own leisure	fixed at 1 (normalisation)
λ_2	child's weight on own skill	the child's own study time
R_0	level of HC productivity	early and late log HC , jointly with investment moments
σ_{10}, σ_{11}	level and age slope of parental-time elasticity	early and late parental time , jointly with HC
σ_{20}, σ_{21}	level and age slope of monetary-investment elasticity	early and late monetary investment , jointly with HC
σ_{40}	log-level parameter of study-time elasticity	early and late child study time , jointly with HC
σ_{41}	age slope of log study-time elasticity	fixed at 0.02
μ_1	slope of the welfare weight	fixed at -0.04 ; see §5

The current SMM estimates **nine parameters against ten moments**. The estimated parameters are $\phi_2, \phi_3, \lambda_2, R_0, \sigma_{10}, \sigma_{11}, \sigma_{20}, \sigma_{21}, \sigma_{40}$. The targets are mean consumption and hours, plus early and late parental time, monetary investment, child study time, and log HC.

These are joint identification arguments, not one-to-one assignments. For example, ϕ_2 affects the leisure–work trade-off, but hours also respond to other parameters. Both valuation and technology can change investment **and therefore resulting HC**. HC moments add information that may help distinguish these channels; they do not, by their presence alone, prove separate identification.

The strength of these arguments depends on matching data and model age coverage and aggregation, and on the residual weights. In particular, dividing a log-HC residual by the target's log level makes its weight depend on the units used to measure HC.

4. For the next step

These are not estimated yet. Listed with candidate informative moments.

Parameter	Role	Candidate informative moments
κ_{term}	weight on the parents' retained assets in their terminal value	parental net worth after the child leaves — family assets at child age 17, and post-separation wealth for families whose child has finished college
κ_0	level of the psychic cost of college	the overall college enrolment rate
κ_θ	ability gradient in the psychic cost (negative)	the enrolment gradient in child test scores — enrolment by achievement quantile
κ_{ParEd}	parental-education shift (negative)	the enrolment gap by parental education

The three psychic-cost parameters all move enrolment. The level, ability gradient, and parental-education gap provide three distinct sources of variation. The overall rate alone cannot regularly identify all three, and three moments alone do not guarantee that their effects are sufficiently independent.

κ_{term} governs the retention–transfer trade-off, making retained wealth a particularly relevant moment. It can also affect enrolment through transfers.

Their present values are transported from Colas (Table 2) by **sign and ratio only** — $\kappa_{\text{ParEd}}/\kappa_\theta = 0.205$ is hers. The levels are not transportable: her utility is $c^{1-\gamma}/(1-\gamma)$ at $\gamma = 1.9$ with consumption in dollars, ours is $\rho = 1.5$ with c of order 0.1–5.

5. Open question: the technology and welfare-weight age profiles

Currently neither μ_1 nor σ_{41} is estimated. Their values are $\mu_1 = -0.04$ and $\sigma_{41} = 0.02$; σ_{40} is estimated. The two slope parameters are calibrated assumptions. The resulting study-time profile is nevertheless endogenous, and its early and late means are already targeted by SMM.

The earlier claim that μ_1 and σ_{41} necessarily enter as a single combination was incorrect. To see the distinction, set $s = t - 5 > 0$. With $\mu_0 = 1$,

$$1 - \tilde{\mu}_t = -\mu_1 s,$$

and the technology-to-leisure-weight ratio is

$$\mathcal{R}_t = \frac{\exp(\sigma_{40} + \sigma_{41}s)}{(-\mu_1)s\lambda_1}.$$

For $\mu_1 < 0$,

$$\log \mathcal{R}_t = \sigma_{40} + \sigma_{41}s - \log(-\mu_1) - \log s - \log \lambda_1.$$

In this ratio, σ_{41} changes the age slope, while μ_1 changes a level term. The level combination instead involves $\sigma_{40} - \log(-\mu_1)$. Even that observation is not a proof of non-identification in the full model: μ_1 also changes the family's HC valuation and its continuation incentives.

For an interior study choice in the logarithmic-leisure region, the condition is

$$\frac{(1 - \tilde{\mu}_t)\lambda_1}{l_{c,t}} = \beta_t \mathbb{E}_t[V_{HC,t+1}] HC_{t+1} \frac{\sigma_{4,t}}{i_{c,t}}.$$

The continuation derivative, resulting HC, and other choices all respond to the parameters. Constraints can also bind. A ratio alone therefore cannot establish which parameters the observed age profile identifies.

The available choices are:

1. **Keep both fixed, as currently implemented.** State their calibrated values explicitly and interpret the estimated parameters conditional on them.
2. **Estimate σ_{41} and retain a stated calibration for μ_1 .** An optional calibration rule $\tilde{\mu}_{17} = 0.5$ gives $\mu_1 = -1/24 \approx -0.0417$; the current value gives $\tilde{\mu}_{17} = 0.52$. This is a proposed rule, not an instruction attributed to Sahber.
3. **Estimate μ_1 and retain a stated calibration for σ_{41} .** The present value is 0.02. Nothing in the early-age zero restriction requires changing it to zero.

Adding either parameter would give ten parameters and ten moments; identification would still need evidence. Adding both while retaining all nine current parameters would give eleven parameters and ten moments, so the moment Jacobian could not have full column rank. That is a restriction of the present moment set, not proof of an inherent equivalence between the two parameters.

No parameter choice or calibration is changed by this memo.

6. Additional robustness exercise from the paper

10/10 — Plot how all estimated parameters respond to each target moment.

Following the identification diagnostics in Sections 4.4 and 5.4 of *structural-robustness-Jan26.pdf*, change one target moment at a time, hold the other targets fixed, and re-estimate **all nine estimated parameters jointly** at every perturbation. Plot each parameter estimate against the perturbed target value. This measures how estimation reallocates a change in the data across parameters.

Use the agreed baseline target definitions and weighting matrix. For each of the ten targets, choose a small range around its baseline, preferably expressed in standard errors of the estimated moment. These are standard errors of the sample moment, not standard deviations of individual observations. If they are unavailable, report the chosen perturbations explicitly without giving them a confidence-interval interpretation.

Keep the weighting matrix, residual scaling, random draws, parameter bounds, and solver settings fixed across perturbations. Reuse the cached child solution because the current nine parameters affect only the parent block. Nearby solutions can supply starting points, but retain alternative starts to check for a different minimum; validate the resulting estimates on the final estimation grid.

Produce ten sets of nine parameter curves, with the baseline target and estimate marked. Record objective values, moment residuals, solver failures, and binding parameter bounds alongside the plots. Large movements, jumps, or boundary solutions flag sensitivity that needs explanation. The fixed μ_1 and σ_{41} are not estimated-parameter curves in the current specification.

The exercise is a conditional sensitivity diagnostic, not a confidence region or proof of global identification. It is proposed here and has not been run. This is the only additional robustness exercise adopted from the paper in this memo.